

BellBot V1 Board

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1 Introduction

The objective of BellBot is to incorporate Simultaneous Localization and Mapping (SLAM) alongside other features in order to achieve luggage transportation typically done by a bellboy. This is a difficult task. Therefore, our team decided to first create a minimum viable product (MVP). This is a prototype that limits the number of features to those necessary to demonstrate the basic functionality of the product. For our team, the following criteria were set for the MVP:

- The product can navigate through a pre-mapped obstacle course using LiDAR
- The product can be remotely controlled using Bluetooth
- The product can store all data from operation in onboard memory for further analysis later
- The product can demonstrate maneuverability in all directions on a 2D plane

In order to achieve the criteria for our MVP, a custom printed circuit board (PCB) was designed. Although it may be possible to accomplish the minimum criteria without the use of a custom PCB by using a breadboard or protoboard, a custom PCB was chosen to provide reliable signal integrity, a cleaner aesthetic, and ease of use.

Our custom PCB uses an STM32 Nucleo-F767ZI development board [1], [2] as the main MCU platform, alongside several commercial off-the-shelf (COTS) through-hole components. These components include an inertial measurement unit (IMU), Light Detection and Ranging (LiDAR) unit, Flash SPI memory card for data storage, HC-05 Bluetooth module, four pulse-width modulation (PWM) 8 V continuous-rotation servo pins, as well as additional Quad-SPI and I2C breakout pins to use as needed. At the end of this section, there are 3D renderings of our board.

Through-hole COTS components were used for the first iteration due to their ease of soldering, ease of use, and proven reliability. Future iterations of this board will switch to an STM32 SMT chip and SMT components to reduce board size and remove unwanted components.

The 3D modeling/CAD was done in tandem with the board development, and those contents are not covered in this document. This document serves as a datasheet/guide for anyone using the BellBot board. The sections that follow are: (2) Power Systems, which describes how power is distributed throughout the board; (3) Peripherals, which describes how the featured peripherals are connected and used by our MCU; (4) Software Configuration, which shows the STM32CubeMX model and pin configurations used; and (5) Future Improvements, which focuses on future design improvements.

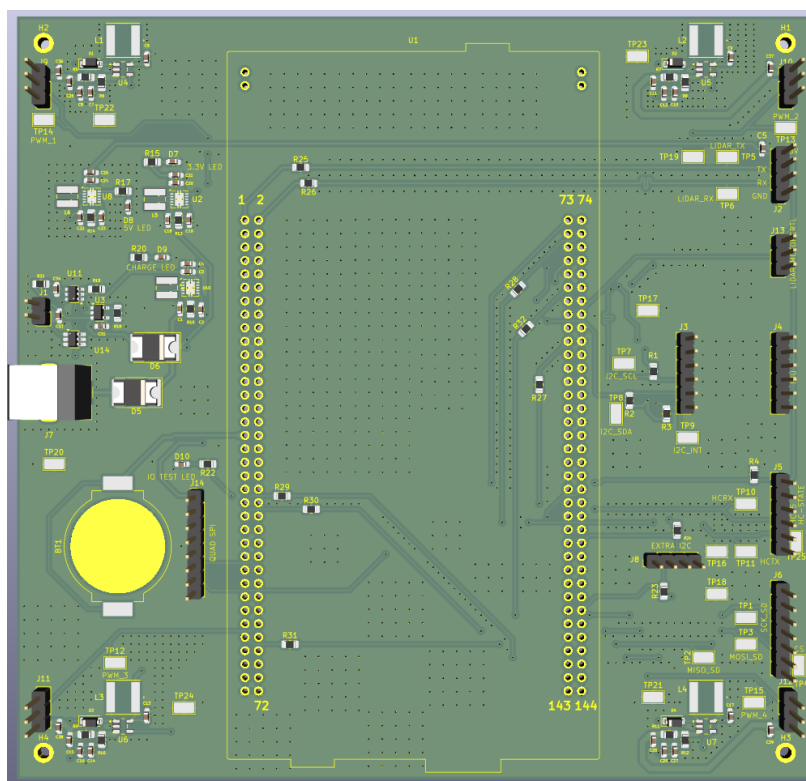


Figure 1: PCB Front View

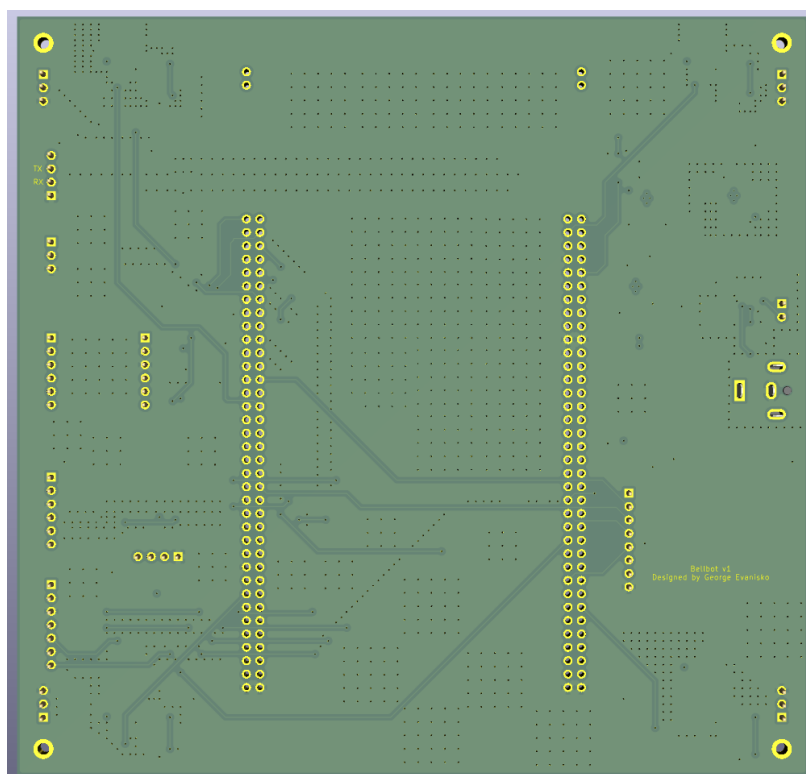


Figure 2: PCB Back View

2 Power Systems

Our board features three different voltage levels: 3.3 V, 5 V, and 8 V. The voltage is initially received from a 5 V, 6 A DC power jack input from a wall plug or from a lithium-ion battery. The board uses a simple diode steering method to choose which source to use by employing Schottky diodes to minimize voltage loss. In addition to source control, onboard chips such as the DW01A (U11) [3], along with dual MOSFET FS8205 (U14) [4] chips, are used for battery protection. The board features a separate 5 V charging regulator (U10) that connects directly to the DC power jack and through a lithium-ion charging chip, the MCP73812T (U3) [5], to charge the battery when the DC power jack is being powered by a source greater than the battery’s nominal voltage of 3.3–3.6 V. This means that when using the 5 V, 6 A DC power jack, the battery will always be in charging mode. A table below describes the charging modes:

Mode	DC Power Jack	Battery	Board Operation
DC powered / battery charging	Connected to 5 V wall supply	Charging through the MCP73812T charging circuit	The DC power jack supplies the board power rails while the battery is charged separately. The diode steering circuit prevents the battery from being the main supply.
Battery powered	Disconnected	Discharging	The lithium-ion battery supplies the board through the diode steering circuit. The 3.3 V, 5 V, and 8 V rails are generated from the battery input as needed.
Battery protection active	Connected or disconnected	Protected by DW01A and FS8205 circuit	The protection circuit disconnects or limits the battery path during unsafe conditions such as over-discharge, overcharge, or overcurrent. Normal board operation may be interrupted until the fault is removed.

Table 1: Battery and power source operating modes

The U10 5V output regulator is used specifically for charging. The U8 5V output regulator is a general use mainly powering the LiDAR unit.

2.1 8V Boost Converter Inductor and Power Calculations

In terms of power regulators, the board features four 5 V to 8 V switching regulators, each used to power a servo motor. The regulator used is an LT1935 and uses a voltage divider with feedback to determine the output using the following equation:

$$V_o = 1.265V * (1 + \frac{R_1}{R_2}) \quad (1)$$

Where R1 and R2 are the resistors used for the feedback voltage divider, as shown by the typical application page:

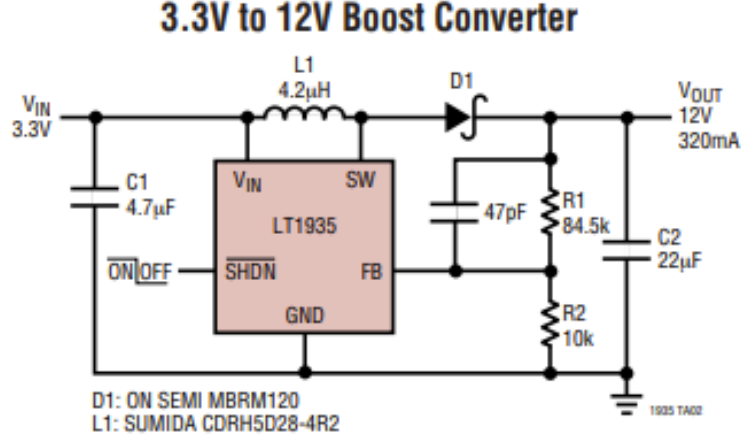


Figure 3: LT1935 typically application [6]

For our usage, $R1 = 56.2K$ and $R2 = 10K$ yields an output voltage of 8.37 V.

The capacitors are arbitrary and any capacitor can be used if it has a low ESR. The inductor value ($L1$) is based on the following formula:

$$L = 3 \left(\frac{V_{IN}}{V_{OUT}} \right) \frac{V_{OUT} - V_{IN}}{I_{MAX} f} \quad (2)$$

Given a maximum switching frequency of 1.2 MHz and an I_{max} of 2 A with 5 V V_{in} and 8 V V_{out} , we can calculate our inductor value as 2.3–2.4 μH , which will be rounded down to 2.2 μH . The output power of the boost converter was also estimated. The output power is related to the input power and converter efficiency, as shown in Equation 3.

$$P_{OUT} = P_{IN} \eta \quad (3)$$

Since input power is equal to input voltage multiplied by input current, this can be written as:

$$P_{OUT} = V_{IN} I_{IN} \eta \quad (4)$$

For this design, the input voltage is 5 V, the maximum input current is approximated using the maximum switch current of 2 A, and the assumed efficiency is 90%.

$$\eta = 0.90 \quad (5)$$

Substituting the values into Equation 4 gives:

$$P_{OUT} = (5 \text{ V})(2 \text{ A})(0.90) \quad (6)$$

$$P_{OUT} = 9 \text{ W} \quad (7)$$

The output current available from the 8 V rail can then be estimated using the basic power relationship shown in Equation 8.

$$I_{OUT} = \frac{P_{OUT}}{V_{OUT}} \quad (8)$$

Substituting the output power and output voltage gives:

$$I_{OUT} = \frac{9 \text{ W}}{8 \text{ V}} \quad (9)$$

$$I_{OUT} = 1.125 \text{ A} \quad (10)$$

Based on this estimate, the 8 V boost converter can supply approximately 9 W of output power, corresponding to about 1.1 A at 8 V when powered from the 5 V input. This value is an estimate because the actual output current will depend on the converter efficiency, thermal performance, current limit behavior, and selected external components. In the event that our device is powered by a 3.3-3.6V li-ion battery our output current would be around 0.75 A.

2.2 5V and 3.3V Buck-Boost Converters

For our 5 V and 3.3 V rails, the TPS60001 and TPS60002 were used. The TPS6000X is a 1.8 A switching buck-boost converter. The TPS60001 and TPS60002 have set output voltages of 3.3 V and 5 V, respectively. Below is a typical application schematic:

8.2 Typical Application

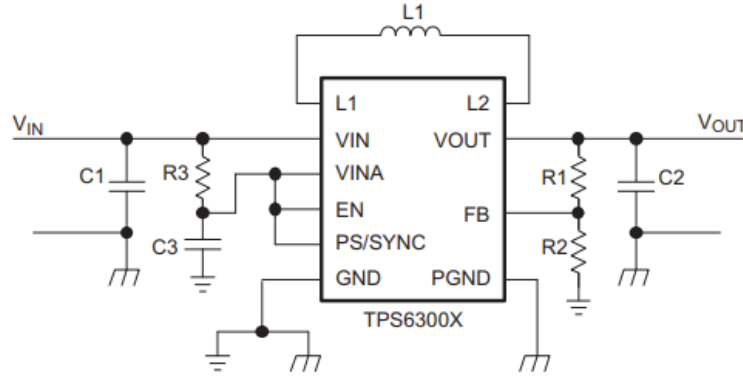


Figure 2. Typical Application Circuit for Adjustable Output Voltage Option

Figure 4: TPS 6000X Typical Application [7]

For the set output voltage versions, TPS60001 and TPS60002, R1 and R2 are not needed, and the FB pin is shorted to Vout. Using the graphs given by TI, with an input voltage of 5 V, we can expect 1600 mA and 1400 mA output current for the TPS60001 and TPS60002, respectively:

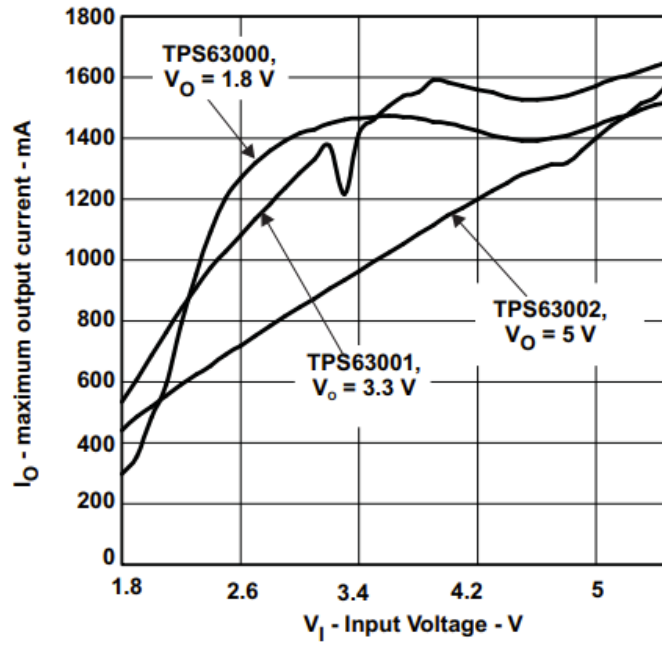


Figure 5: TPS 6000X Output Voltage [7]

At 3.4 V, which is the approximate voltage level for a lithium-ion battery, we can expect output currents of 1400 mA and 950 mA for the TPS60001 and TPS60002, respectively.

2.3 Power Subsystem Bill of Materials

Table 2: Major power subsystem components

Designator	Part Number	Use
U4, U5, U6, U7	LT1935	Boost switching regulator used to generate the 8 V rail from the 5 V input. Each LT1935 regulator powers one servo motor.
U2	TPS60001	Buck-boost switching regulator used to generate the regulated 3.3 V rail. This rail can be maintained when powered from either the 5 V DC input or the lithium-ion battery.
U8, U10	TPS60002	Buck-boost switching regulator used to generate the regulated 5 V rail. This rail can be maintained when powered from either the DC input or the lithium-ion battery.
U3	MCP73812T	Lithium-ion battery charging IC used to charge the battery when the external DC power jack is connected.
U11	DW01A	Lithium-ion battery protection IC used to monitor the battery and protect against unsafe conditions such as overcharge, over-discharge, and overcurrent.
U14	FS8205	Dual MOSFET used with the DW01A battery protection IC to disconnect or protect the battery path during fault conditions.
D5, D6	Schottky Diode	Used for diode steering between the external DC power jack and the lithium-ion battery. The Schottky diode is used to reduce voltage drop while preventing reverse current between power sources.
J7	5 V, 6 A DC Power Jack	Main external power input for the board. When connected, it powers the board and allows the battery charging circuit to charge the lithium-ion battery.
J1	Lithium-Ion Battery Connector	Backup or portable power source for the board when the external DC power jack is disconnected.
BT1	Back-up Coin Cell Emergency Power	Backup 3V coin cell battery to preserve data in case of an emergency/unexpected shutdown.

3 Peripherals

Our board has four onboard peripheral components: an IMU, LiDAR, memory card, and Bluetooth module. It also has breakout pins for an additional Quad-SPI and I2C peripheral in the event they are needed. There are also four PWM pins, one in each corner, used to control four servo motors. Due to the long length of our transmission lines, 22 Ohm resistors were added in series at the start of each transmission line to prevent ringing.

3.1 IMU

Our board uses a through-hole COTS IMU from Adafruit with 9 degrees of freedom (DOF) [8]. This component can communicate over SPI or I2C. I2C was chosen because it uses fewer lines than SPI. This sensor takes a 3.3 V–5 V input and has an active-low reset. It consumes 7.5 mA when active. The functionality of the IMU is listed below:

- **Acceleration Vector / Accelerometer:** Returns three-axis acceleration data, including both gravity and linear motion, in m/s^2 .
- **Angular Velocity Vector / Gyroscope:** Returns three-axis rotational speed data in rad/s .

- **Magnetic Field Strength Vector / Magnetometer:** Returns three-axis magnetic field measurements in microtesla (μT).
- **Linear Acceleration Vector:** Returns three-axis linear acceleration data with the gravity component removed, in m/s^2 .
- **Gravity Vector:** Returns three-axis gravitational acceleration data with linear movement removed, in m/s^2 .
- **Absolute Orientation / Rotation Vector:** Returns a four-point quaternion output that can be used for accurate orientation tracking and data manipulation.

3.2 LiDAR

Our board uses a LiDAR kit from Adafruit, specifically the SLAMTEC RP-LiDAR A1 [9]. This kit communicates via UART, takes 5 V for both LiDAR and servo motor operation, and consumes approximately 600 mA on startup and 450 mA after that. The motor controller uses a GPIO pin to enable motor operation. The pins on the LiDAR do not match one-to-one with the pins on our board. For example, TX and RX are in positions 1 and 2 for the kit but positions 2 and 3 on our board. Please take note of this when setting up the LiDAR. The TX from the board should connect to the RX on the LiDAR and vice versa. Some key features of our LiDAR kit are:

- Provides 360 degree two-dimensional scanning, allowing the system to generate point cloud data for mapping, localization, and obstacle detection.
- Supports a maximum measurement range of up to 12 m for the A1M8-R5 model, with a reduced 6 m range for some earlier A1M8 models.
- Uses laser triangulation to measure distance and outputs both distance and angle data for each sampled point.
- Has a typical scan rate of 5.5 Hz and can be configured up to a maximum scan rate of 10 Hz.
- Supports a sampling frequency of at least 8000 samples per second, allowing the LiDAR to collect dense scan data during operation.
- Outputs sample point data that includes distance, heading angle, measurement quality, and a start flag indicating the beginning of a new scan.
- Uses a low-power infrared laser and is rated as a Class I laser device, making it safe for normal operation when used as specified.
- Includes built-in speed detection and adaptive control, allowing the scanner system to adjust based on the motor speed.
- Includes self-protection and status detection features that can shut down the laser if unsafe or abnormal operating conditions are detected.

The RPLIDAR A1 outputs continuous sample point data frames through its communication interface. Each sample point contains the measured distance in millimeters, the heading angle in degrees, a measurement quality value, and a start flag indicating the beginning of a new scan [9].

Data Field	Unit	Description
Distance	mm	Measured distance between the rotating core of the RPLIDAR A1 and the sampled point.
Heading	Degrees	Current heading angle of the measurement.
Quality Level	Unitless	Indicates the quality of the distance measurement.
Start Flag	Boolean	Indicates the start of a new 360 degree scan.

Table 3: RPLIDAR A1 sample point data format

Below is a picture of the LiDAR controls for reference:

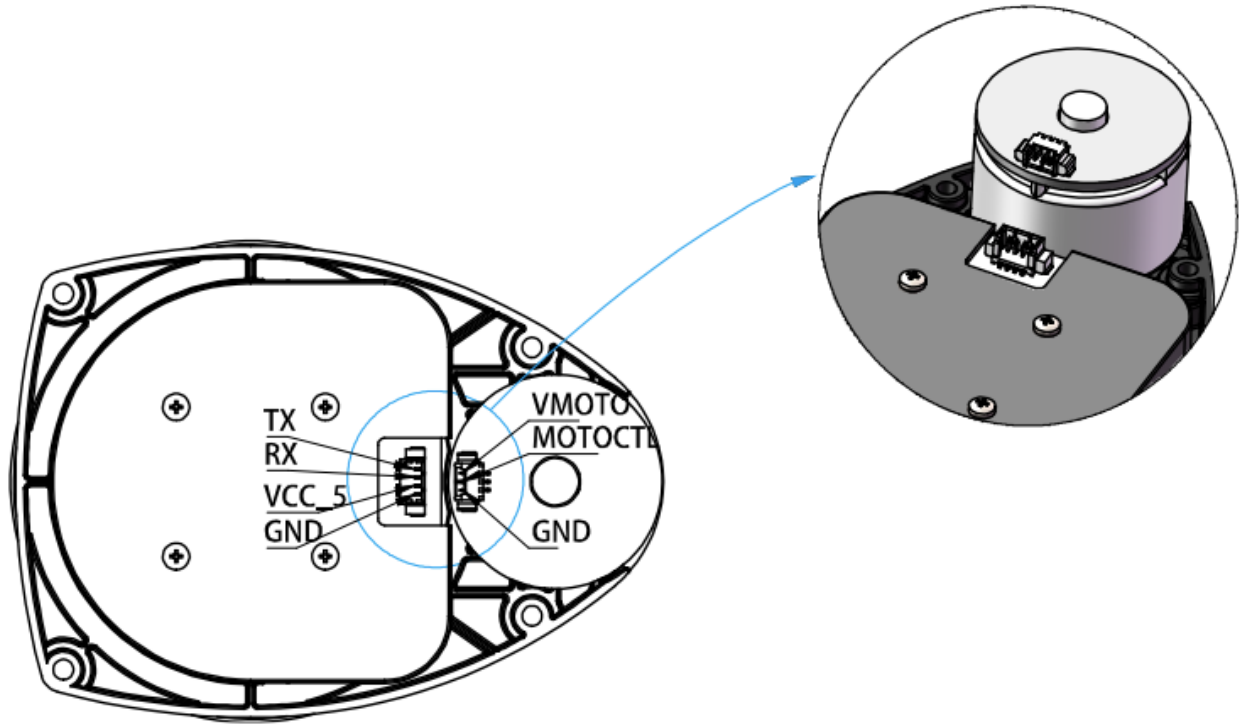


Figure 2-4 RPLIDAR A1 Power Interface

Figure 6: LIDAR Pinout [9]

3.3 SD Flash SPI Memory Card

Our board uses a through-hole COTS SPI Flash SD Card breakout from Adafruit for data storage [10]. The module is designed to behave like an SD card in an SPI flash package, allowing it to be used with standard SD card libraries. This makes it useful for storing sensor data, debug logs, or other information collected by the MCU during operation. This module takes a 3.3 V–5 V input voltage and consumes around 100 mA of current when active.

The SPI Flash SD Card communicates over SPI, using MOSI, MISO, SCK, and CS lines. The breakout supports both 3.3 V and 5 V logic through onboard level shifting and includes a 3.3 V regulator. The CS pin also includes a pull-up resistor, which helps keep the device deselected when the MCU is not actively communicating with it [10]. Some key features of the SPI Flash SD Card are listed below:

- Provides non-volatile data storage, allowing the system to save data even when power is removed.
- Communicates using the SPI interface, requiring MOSI, MISO, SCK, and CS connections to the MCU.
- Can be used with standard SD card libraries, simplifying software integration.
- Supports both 3.3 V and 5 V logic through onboard level shifting.
- Includes an onboard 3.3 V regulator, allowing the module to be powered from the board's regulated supply.
- Includes a pull-up resistor on the CS pin, helping prevent accidental selection of the device during startup or reset.
- Can be used for storing sensor readings, LiDAR data, IMU data, system logs, or other information generated during operation.

3.4 Bluetooth Module

Our board uses a COTS HC-05 Bluetooth module to provide wireless serial communication between the MCU and an external device [11]. The HC-05 communicates using UART, it is powered using 5V but its TX and RX lines are 3.3V TTL. The device consumes about 30mA when active [11].

Some key features of the HC-05 Bluetooth module are listed below:

- Provides wireless serial communication using Bluetooth.
- Can be used to transmit sensor data, system status information, or debug messages to an external device.
- Supports simple point-to-point communication, making it useful for basic wireless control or monitoring.
- Uses a small number of signal lines, requiring only UART TX, UART RX, power, and ground for normal operation.
- Can be configured using AT commands, allowing settings such as the module name, baud rate, and pairing password to be changed.
- Allows the board to send data wirelessly without requiring a USB cable or direct serial connection during testing.

3.5 Peripherals Bill of Materials

Designator	Part Number	Use
U1	STM32 Nucleo-F767ZI	Main MCU development board used to control the system, communicate with peripherals, generate PWM signals, and process sensor data.
J3, J4	BNO085	9-DOF IMU used to measure acceleration, angular velocity, magnetic field strength, linear acceleration, gravity, and absolute orientation.
J2, J13	SLAMTEC RPLIDAR A1	360 degree LiDAR sensor used for distance scanning, obstacle detection, mapping, and localization.
J6	Adafruit SPI Flash SD Card	SPI-based storage module used to save sensor data, LiDAR data, IMU data, debug logs, or other system information.
J5	HC-05	Bluetooth serial module used for wireless UART communication between the MCU and an external Bluetooth device.
J9, J10, J11, J12	Servo PWM Header	Header pins used to provide PWM control signals to the four continuous-rotation servo motors.
J8	Quad-SPI Breakout Header	Expansion header used to provide access to Quad-SPI signals for future peripherals or memory expansion.
J14	I2C Breakout Header	Expansion header used to provide access to I2C communication lines for additional sensors or peripherals.

Table 4: Major MCU and peripheral subsystem components

4 Software Configuration

Below is the a screenshot from the IOC file being used for this project in CubeMX. The IOC file is included in the GitHub in the STM32CubeData folder.

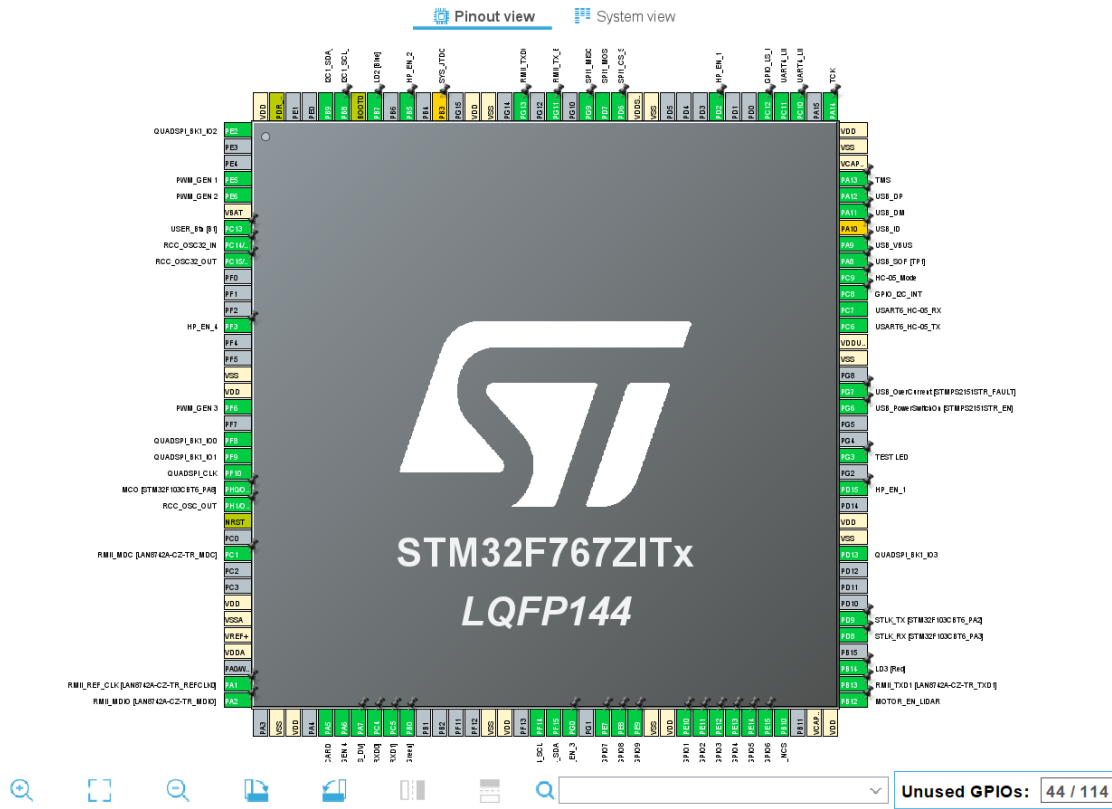


Figure 7: CubeMX IOC Rendering

5 Future Improvements

Future improvements mainly involve a reduction in size by eliminating COTS through-hole components with SMT components. The STM32-F767IZ Nucleo board will be replaced with an STM32 N6 or equivalent QFP/BGA chip with an AI accelerator. The COTS peripherals will be replaced with SMT equivalents. Further improvements will be determined experimentally when the board is assembled and field tested.

One error was found in the board which is the reset pin on the IMU is left floating when it should be pulled-up to high to prevent floating reset (active-LOW reset).

Chip selection (MCU) in future versions will focus on AI acceleration on MCU (using models such as STM32ModelZoo or MCUNet) [12].

References

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